tf.compat.v1.distributions.Multinomial Stay organized with collections Save and categorize content based on your preferences.

https://www.tensorflow.org/api_docs/python/tf/compat/v1/distributions/Multinomial

> *Multinomial distribution.*

Function Signatures

```
tf.compat.v1.distributions.Multinomial( total_count,
logits=None, probs=None, validate_args=False,
allow_nan_stats=True, name='Multinomial' )
```

Code Examples

```
tf.compat.v1.distributions.Multinomial(
total_count,
logits=None,
probs=None,
validate_args=False,
allow_nan_stats=True,
name='Multinomial'
)
```

```
tf.compat.v1.distributions.Multinomial(
total_count,
logits=None,
probs=None,
validate_args=False,
allow_nan_stats=True,
name='Multinomial'
)
```

```
pmf(n; pi, N) = prod_j (pi_j)**n_j / Z
Z = (prod_j n_j!) / N!
```


```
pmf(n; pi, N) = prod_j (pi_j)**n_j / Z
Z = (prod_j n_j!) / N!
```


```
K <= min(2**31-1, {
tf.float16: 2**11,
tf.float32: 2**24,
tf.float64: 2**53 }[param.dtype])
```


```
K <= min(2**31-1, {
tf.float16: 2**11,
tf.float32: 2**24,
tf.float64: 2**53 }[param.dtype])
```


```
logits = [-50., -43, 0]
dist = Multinomial(total_count=4., logits=logits)
```


```
logits = [-50., -43, 0]
dist = Multinomial(total_count=4., logits=logits)
```


Documentation

Multinomial distribution.

Inherits From: Distribution

This Multinomial distribution is parameterized by probs, a (batch of) length-K prob (probability) vectors (K > 1) such that tf.reduce_sum(probs, -1) = 1, and a total_count number of trials, i.e., the number of trials per draw from the Multinomial. It is defined over a (batch of) length-K vector counts such that tf.reduce_sum(counts, -1) = total_count. The Multinomial is identically the Binomial distribution when K = 2.

Mathematical Details

The Multinomial is a distribution over K-class counts, i.e., a length-K vector of non-negative integer counts = n = [n_0, ..., n_{K-1}].

The probability mass function (pmf) is,

where:

- probs = pi = [pi_0, ..., pi_{K-1}], pi_j > 0, sum_j pi_j = 1,
- total_count = N, N a positive integer,
- Z is the normalization constant, and,
- N! denotes N factorial.

Distribution parameters are automatically broadcast in all functions; see examples for details.

Pitfalls

The number of classes, K, must not exceed:

- the largest integer representable by self.dtype, i.e.,

2**(mantissa_bits+1) (IEE754),

- the maximum Tensor index, i.e., 2**31-1.

In other words,

Examples

Create a 3-class distribution, with the 3rd class is most likely to be drawn, using logits.

Create a 3-class distribution, with the 3rd class is most likely to be drawn.

The distribution functions can be evaluated on counts.

Create a 2-batch of 3-class distributions.

Attributes

Stats return +/- infinity when it makes sense. E.g., the variance of a Cauchy distribution is infinity. However, sometimes the statistic is undefined, e.g., if a distribution's pdf does not achieve a maximum within the support of the distribution, the mode is undefined. If the mean is undefined, then by definition the variance is undefined. E.g. the mean for Student's T for df = 1 is undefined (no clear way to say it is either + or -

infinity), so the variance = E[(X - mean)**2] is also undefined.

May be partially defined or unknown.

The batch dimensions are indexes into independent, non-identical parameterizations of this distribution.

May be partially defined or unknown.

Currently this is one of the static instances
distributions.FULLY_REPARAMETERIZED
or distributions.NOT_REPARAMETERIZED.

Methods

batch_shape_tensor

View source

Shape of a single sample from a single event index as a 1-D Tensor.

The batch dimensions are indexes into independent, non-identical parameterizations of this distribution.

View source

Cumulative distribution function.

Given random variable X, the cumulative distribution function cdf is:

View source

Creates a deep copy of the distribution.

covariance

View source

Covariance.

Covariance is (possibly) defined only for non-scalar-event distributions.

For example, for a length-k, vector-valued distribution, it is calculated as,

where Cov is a (batch of) k x k matrix, 0 <= (i, j) < k, and E denotes expectation.

Alternatively, for non-vector, multivariate distributions (e.g., matrix-valued, Wishart), Covariance shall return a (batch of) matrices under some vectorization of the events, i.e.,

where Cov is a (batch of) k' x k' matrices,
0 <= (i, j) < k' = reduce_prod(event_shape), and Vec is some function mapping indices of this distribution's event dimensions to indices of a length-k' vector.

cross_entropy

View source

Computes the (Shannon) cross entropy.

Denote this distribution (self) by P and the other distribution by Q. Assuming P, Q are absolutely continuous with respect to one another and permit densities p(x) dr(x) and q(x) dr(x), (Shanon) cross entropy is defined as:

where F denotes the support of the random variable X ~ P.

entropy

View source

Shannon entropy in nats.

event_shape_tensor

View source

Shape of a single sample from a single batch as a 1-D int32 Tensor.

is_scalar_batch

View source

Indicates that batch_shape == [].

is_scalar_event

View source

Indicates that event_shape == [].

kl_divergence

View source

Computes the Kullback--Leibler divergence.

Denote this distribution (self) by p and the other distribution by q. Assuming p, q are absolutely continuous with respect to reference measure r, the KL divergence is defined as:

where F denotes the support of the random variable X ~ p, H[., .] denotes (Shanon) cross entropy, and H[.] denotes (Shanon) entropy.

log_cdf

View source

Log cumulative distribution function.

Given random variable X, the cumulative distribution function cdf is:

Often, a numerical approximation can be used for log_cdf(x) that yields a more accurate answer than simply taking the logarithm of the cdf when x << -1.

log_prob

View source

Log probability density/mass function.

Additional documentation from Multinomial:

For each batch of counts, value = [n_0, ...
,n_{k-1}], P[value] is the probability that after sampling self.total_count

draws from this Multinomial distribution, the number of draws falling in class j is n_j. Since this definition is exchangeable; different sequences have the same counts so the probability includes a combinatorial coefficient.

log_survival_function

View source

Log survival function.

Given random variable X, the survival function is defined:

Typically, different numerical approximations can be used for the log survival function, which are more accurate than 1 - cdf(x) when x >> 1.

View source

View source

param_shapes

View source

Shapes of parameters given the desired shape of a call to sample().

This is a class method that describes what key/value arguments are required to instantiate the given Distribution so that a particular shape is returned for that instance's call to sample().

Subclasses should override class method _param_shapes.

param_static_shapes

View source

param_shapes with static (i.e. TensorShape) shapes.

This is a class method that describes what key/value arguments are required to instantiate the given Distribution so that a particular shape is
return